Centre for Semiconductor Technologies (SEMIX)

ELECTRICAL ENGINEERING

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EE724
Nanoelectronics Course Projects
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Optimum Device Design for NIPIN Selector

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Outline

- Understand Impact Ionization
- Analyse
  - PIN device
  - NIPIN device
    - Uniform charge distribution
    - Delta charge distribution
- Experiment NIPIN device for
  - Effect of varying lengths of i1 and i2 regions
  - Effect of temperature on IV characteristics
  - Effect of proportion of Germanium in p-region

Motivation

- Advantages of NIPIN selector over NPN selector
  - NIPIN : linear band diagram, NPN : non-linear band diagram
  - NIPIN : ideal subthreshold slope and negligible degradation, NPN : subthreshold slope degradation at high V
- Experimental demonstration of impact ionization

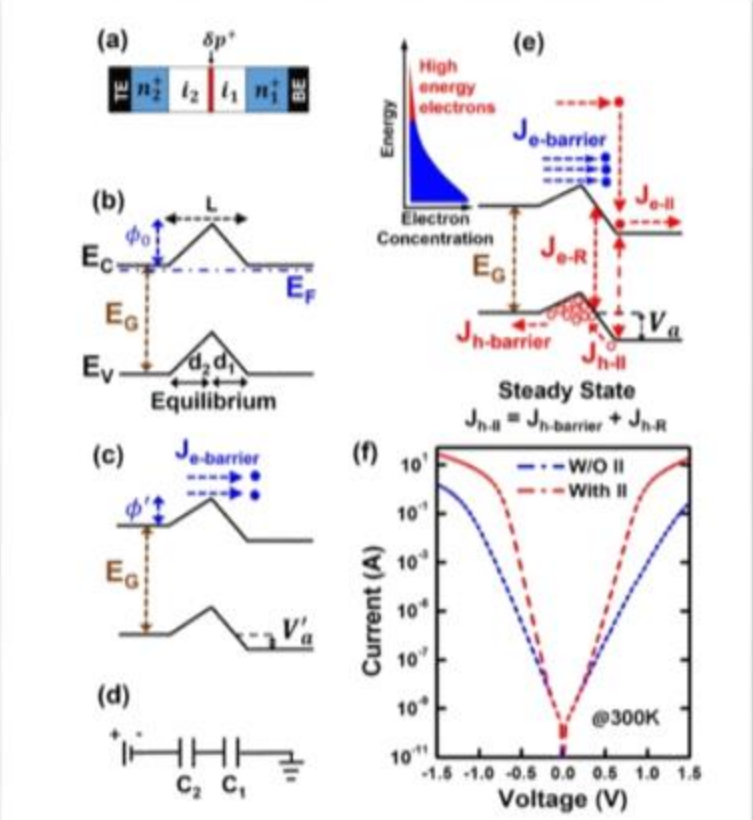


Fig. 1. (a) Schematic of the n-i-p-i-n diode. (b) Equilibrium band diagram of fully depleted punch-through n-i-p-i-n diode has triangular barrier with equilibrium barrier height (ϕ_0). (c) Steady-state band diagram at lower applied bias V_a' without II along with. (d) Simple capacitive voltage divider model for barrier height modulation by applied bias (V_a'). (e) Steady-state band diagram at higher V_a with II. In the BD (electron concentration versus energy) of electrons being injected, most electrons (solid blue circle) in the e current do not have sufficient energy for II but impact ionizing electrons (e_{II}) in high energy "tail" (red circle) can impact ionize to generate holes (open red circle). (f) Comparison of drift diffusion simulated IV characteristics for n-i-p-i-n diode at 300 K with (red dashed line) and without II (blue dashed line).

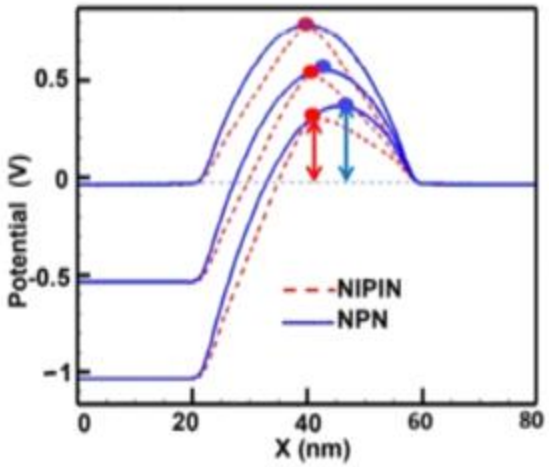
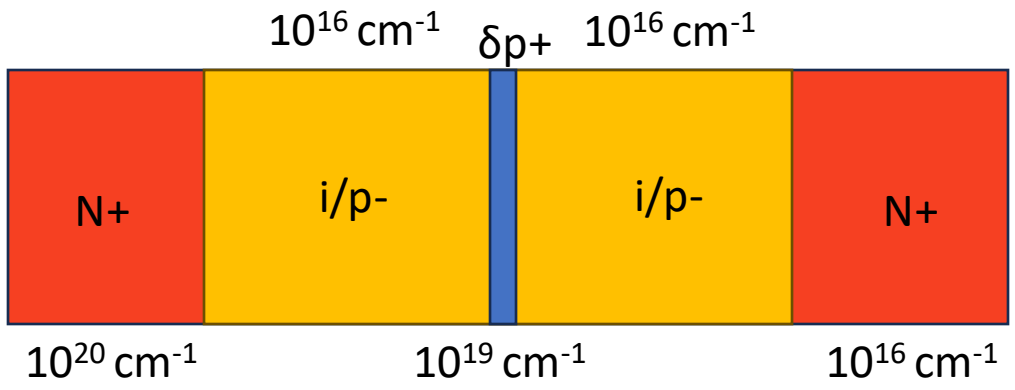


Fig.3: TCAD based conduction bands of NPN and NIPIN selector for different applied voltages. We observe that barrier position is effectively unchanged and barrier height is modulated strongly for NIPIN compared to NPN.

High current, sub ±2V operation, voltage asymmetry

Device Details



- Device consists of 5 regions : n+, i1, p+, i2, n+
- i1 are i2 are p-regions but relatively intrinsic w.r.t n+ and p+
- Device is symmetric/asymmetric based on if i1 and i2 have equal lengths and doping concentration
- Length of p region is very small (~1 nm) as compared to other regions (~10nm)
- p-region is delta-doped

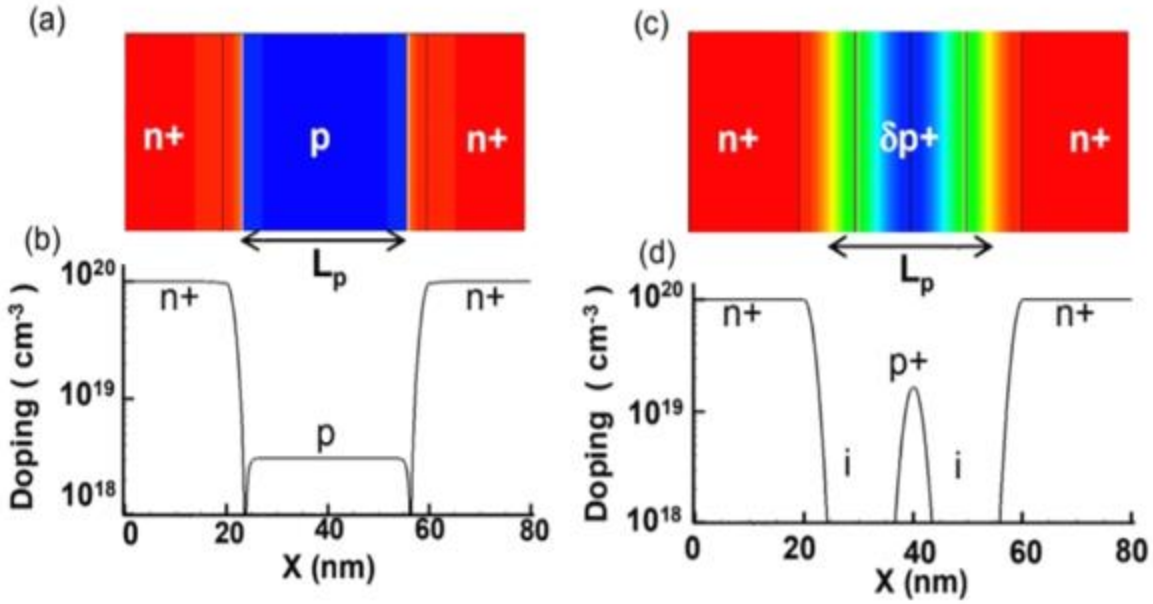


Fig. 1. (a) Structure of conventional NPN selector, (b) NPN doping profile. (c) Structure of NIPIN selector and (d) δ-doped p+ layer surrounded by i-layers for NIPIN selector.

Highly p- doped intrinsic region and δ-doped p+ region

TCAD calibration

Si-Ge in p-region:

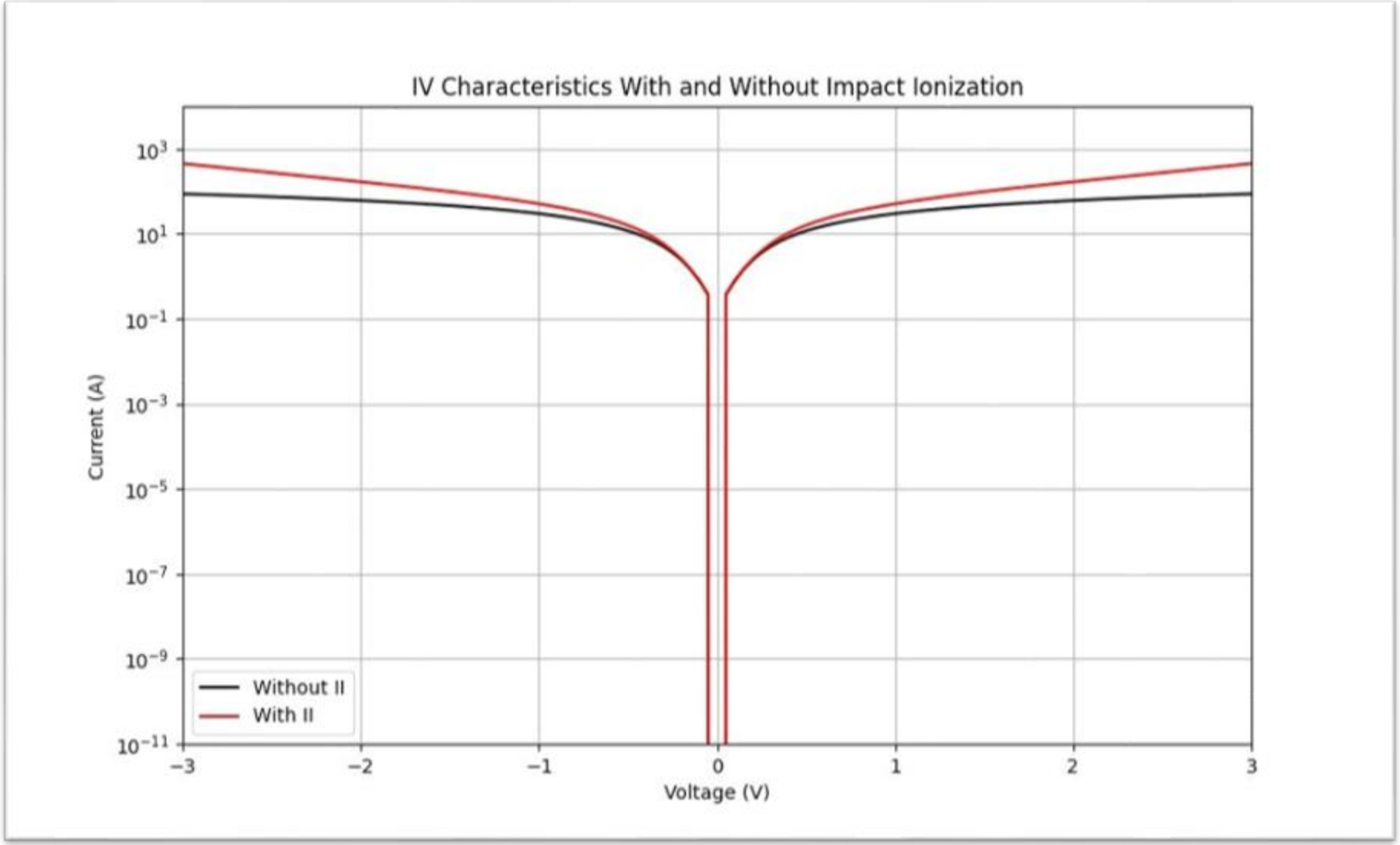
```
(sdegeo:create-rectangle
        (position xminp yminp 0) (position xmaxp ymaxp 0)
        "SiliconGermanium" "p"
)
```

```
HeteroInterface
MoleFraction( RegionName = ["p"]
xFraction = 1.00)
```

Impact Ionization:

```
eAvalanche
hAvalanche
```

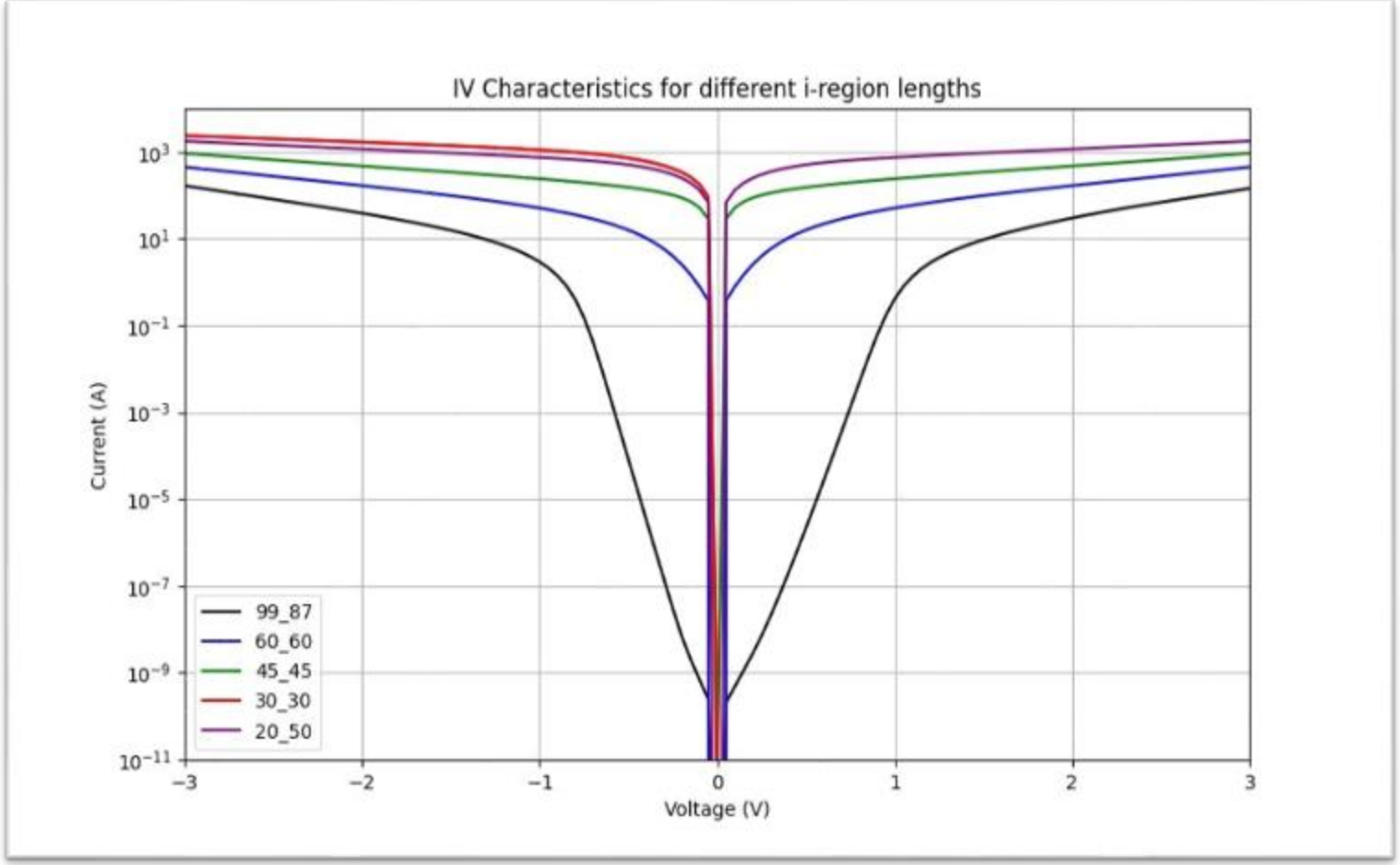
Impact Ionization



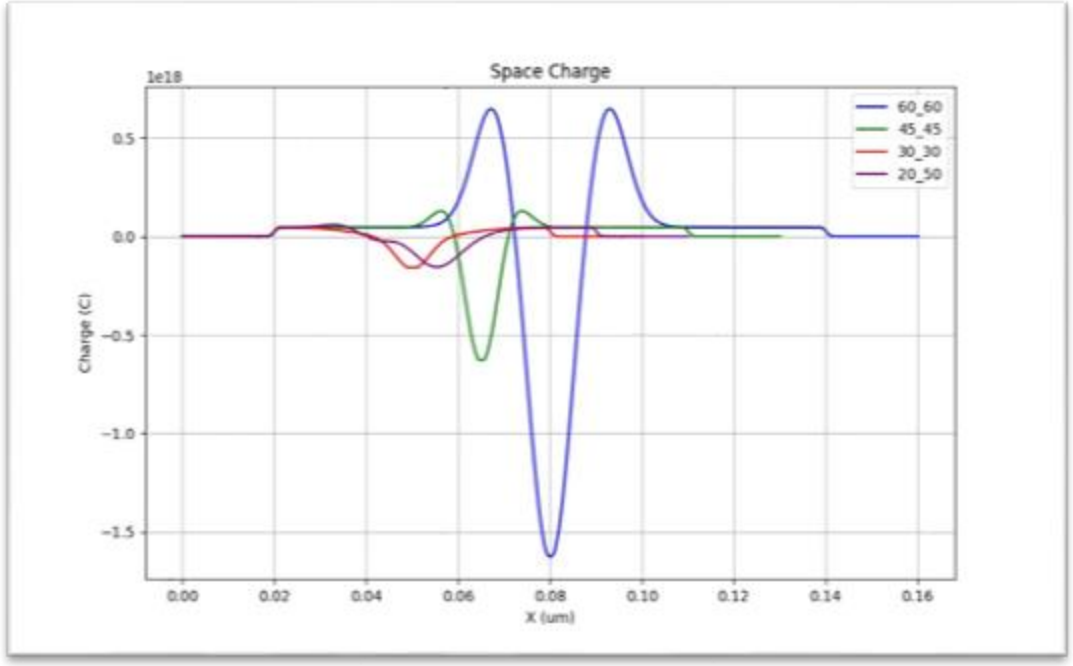
Impact of parameter variation

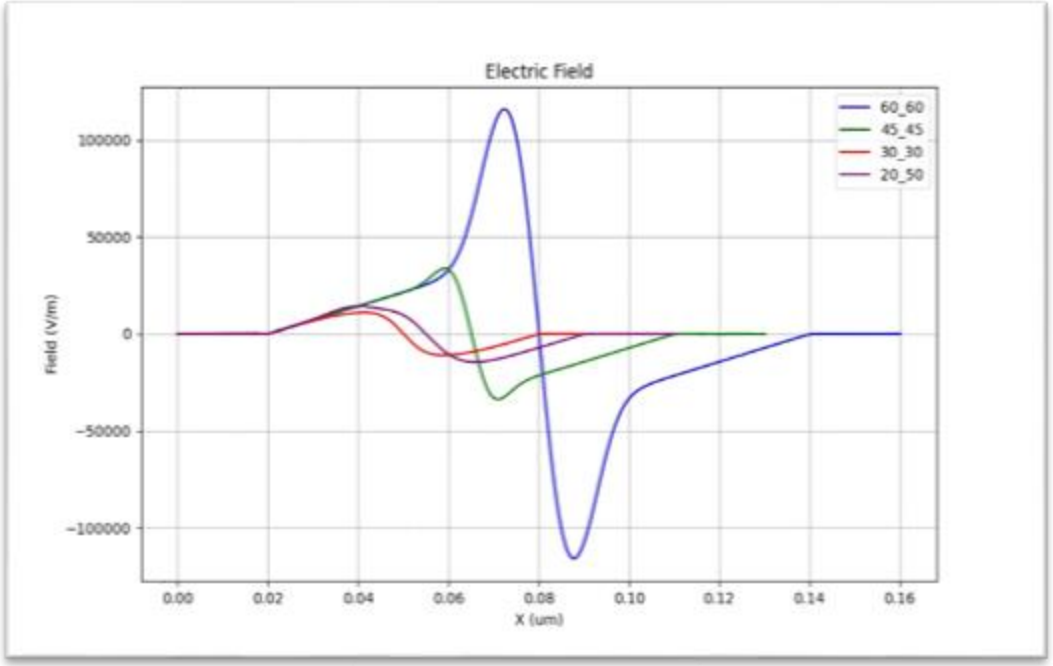
Length of i1 and i2 regions

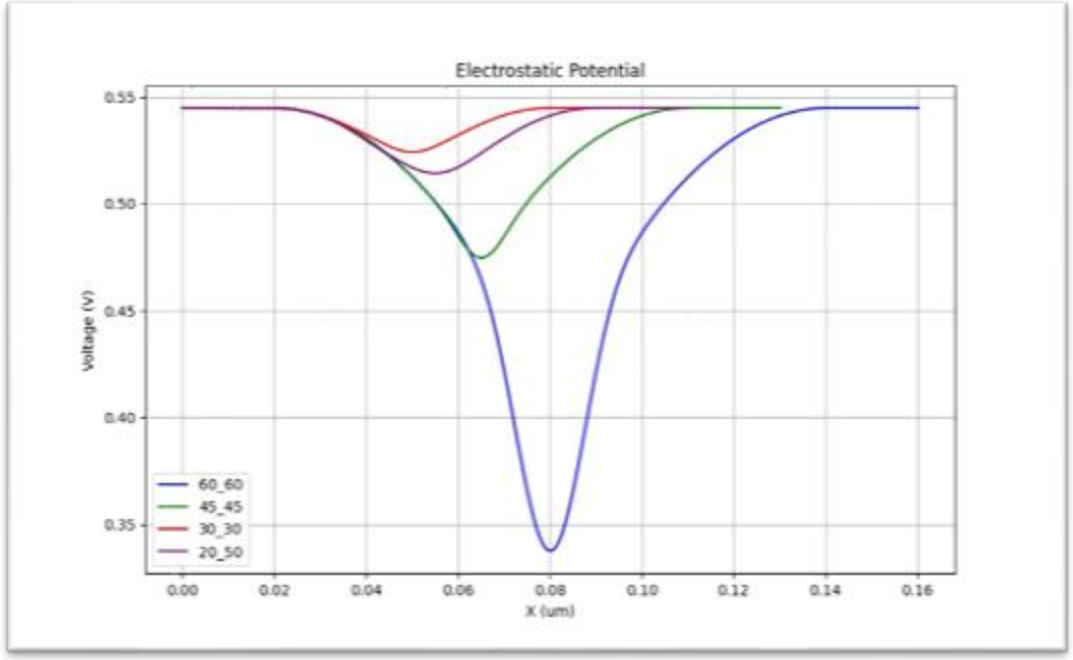
- Device is symmetric/asymmetric if length of i1, i2 are equal/unequal respectively
- IV characteristics/Band diagram are symmetric/asymmetric likewise

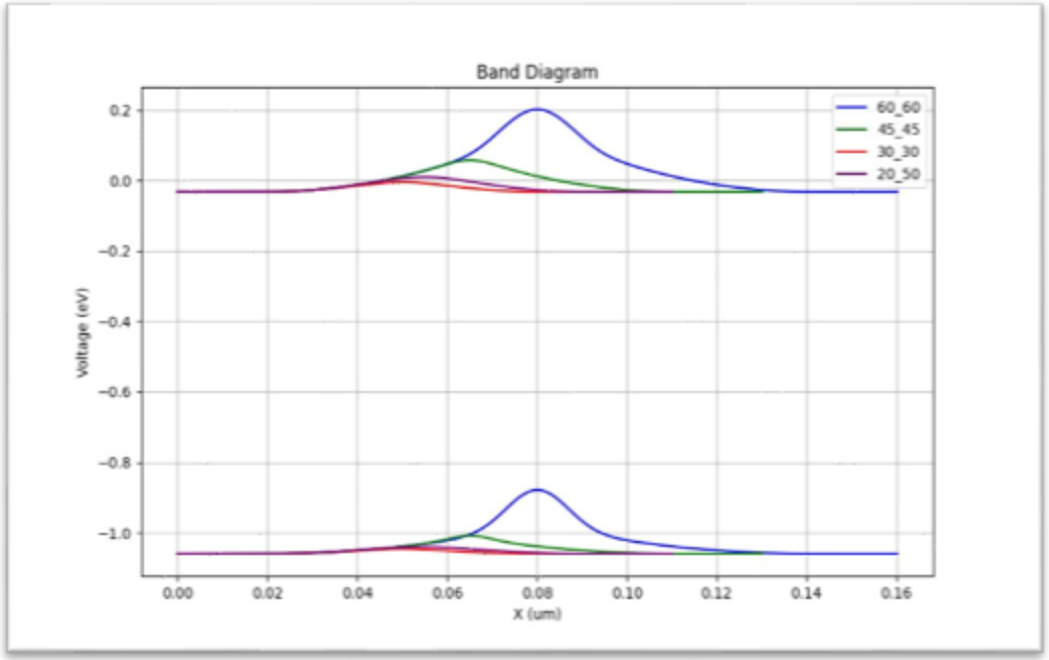


Symmetric and asymmetric IV characteristics


Space Charge
1e18
Charge (C)
X (um)
60_60
45_45
30_30
20_50

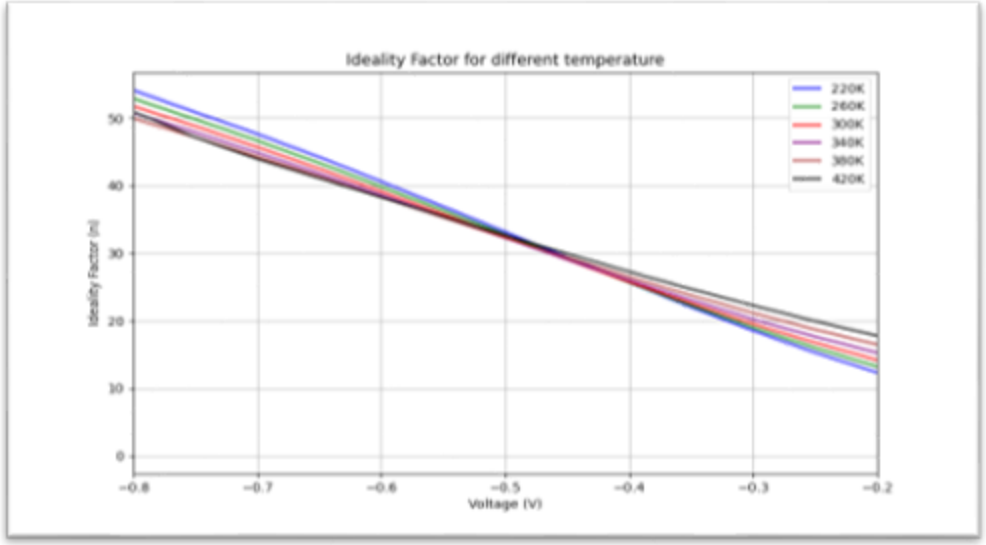

Electric Field
Field (V/m)
X (um)
60_60
45_45
30_30
20_50

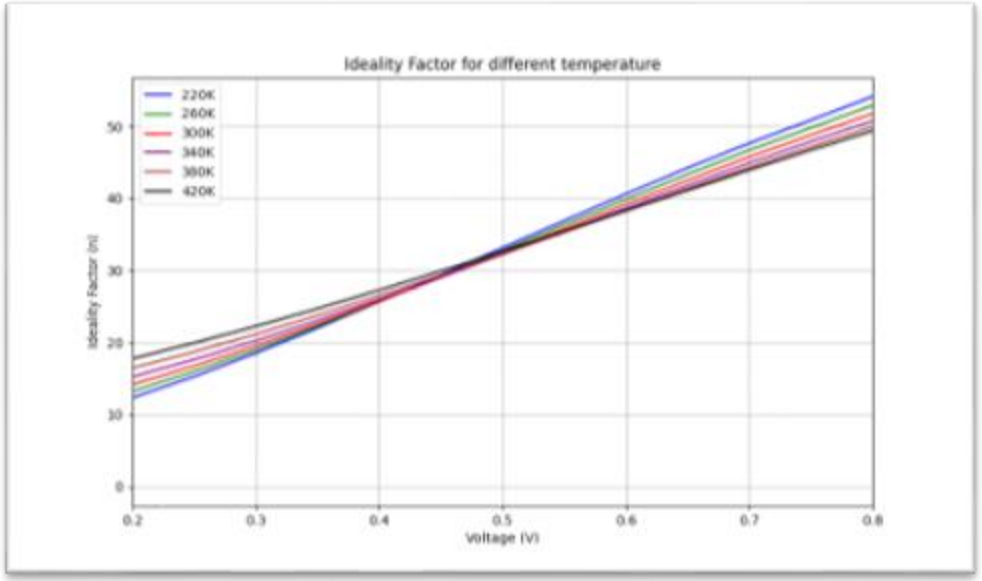

Electrostatic Potential
Voltage (V)
X (um)
60_60
45_45
30_30
20_50

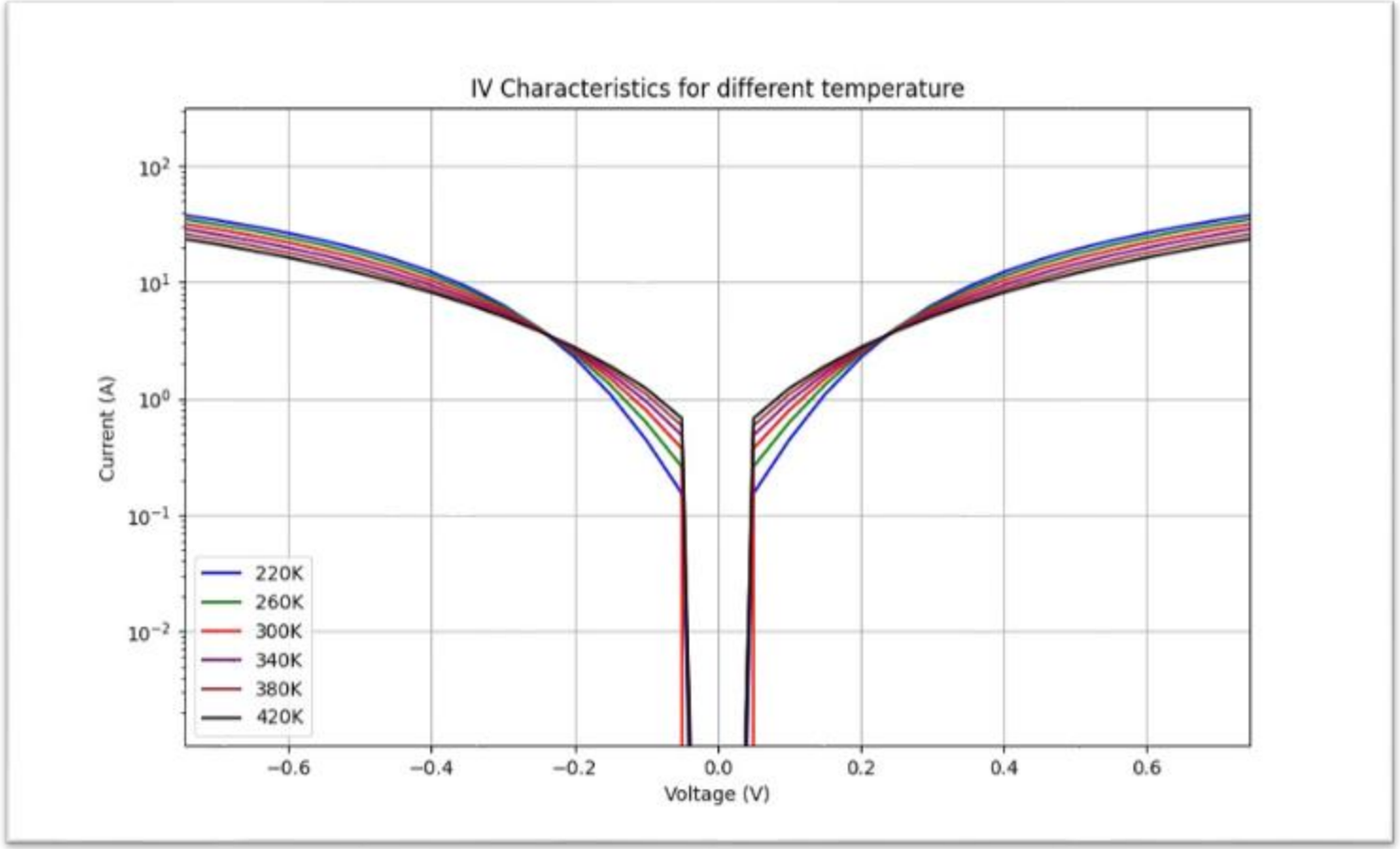

Band Diagram
Voltage (eV)
X (um)
60_60
45_45
30_30
20_50

Impact of parameter variation

Temperature of operation



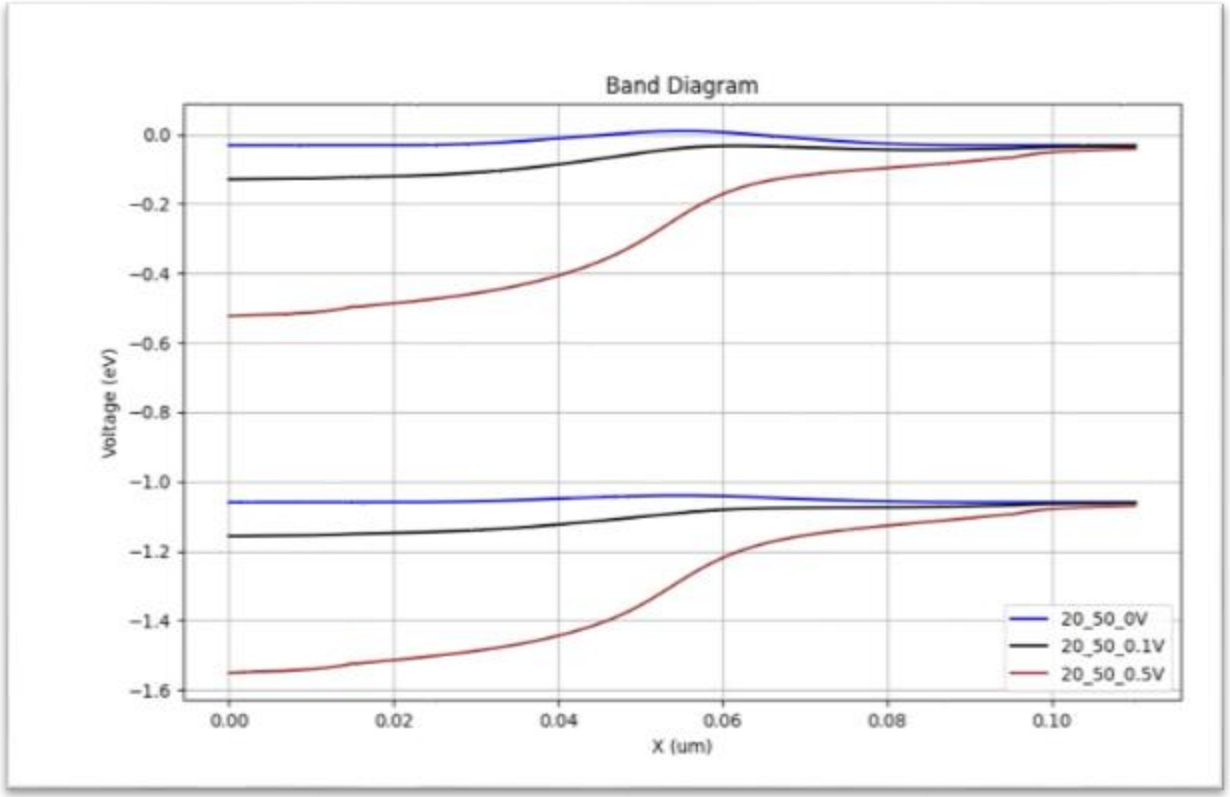


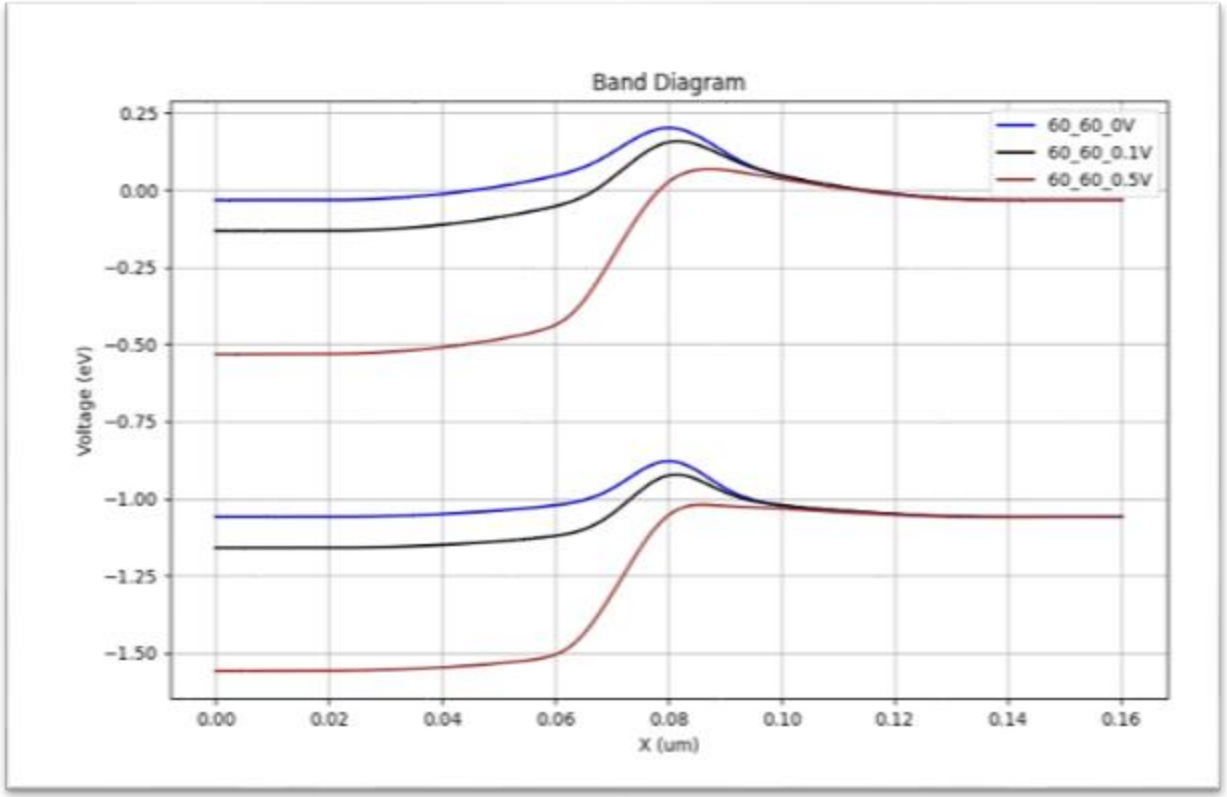


Existence of cross over point

Impact of parameter variation

Bias voltages

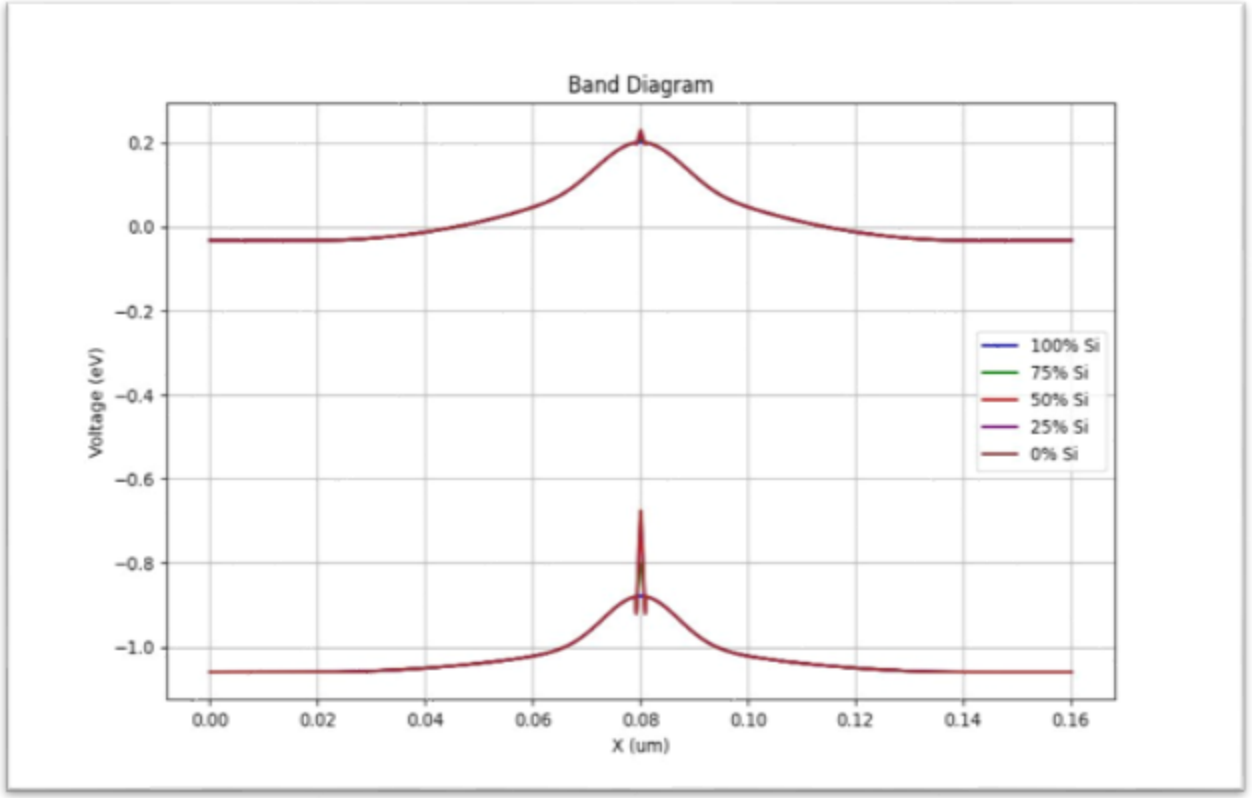


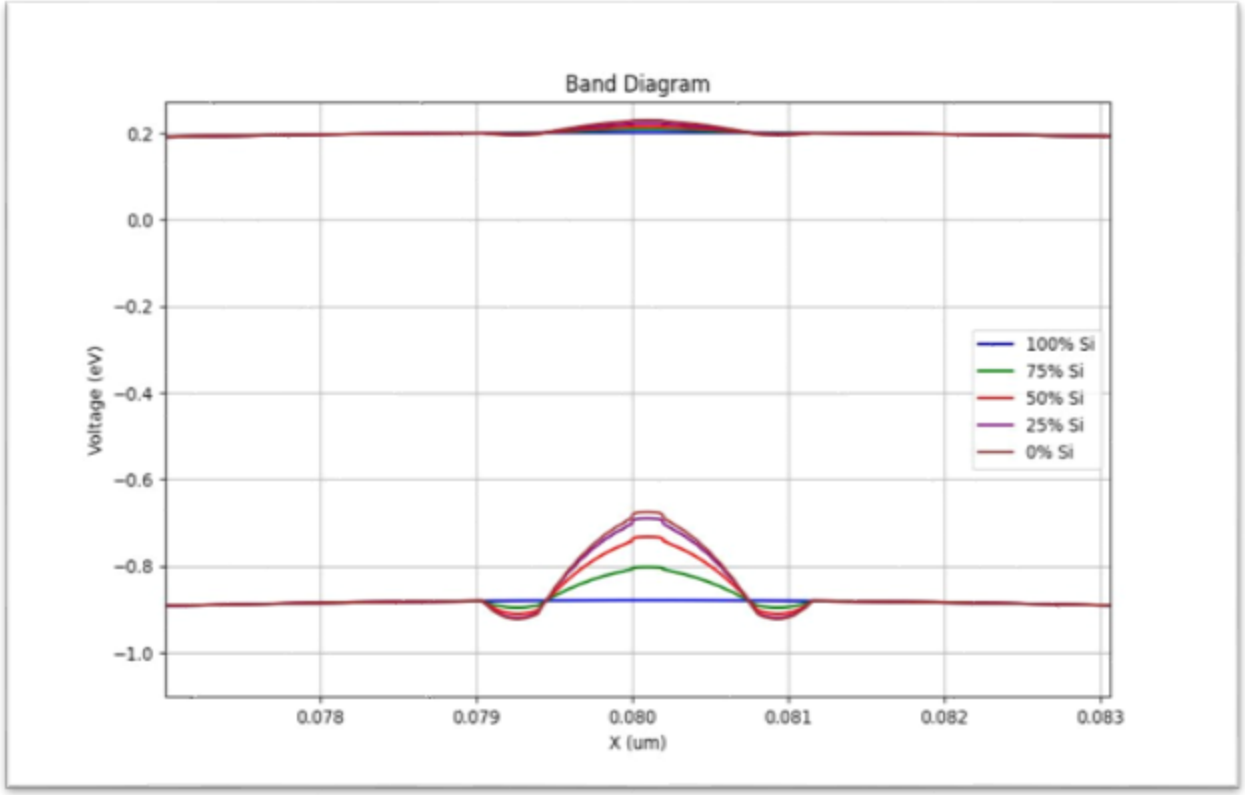


Barrier height increases with positive bias

Impact of parameter variation

Proportion of Germanium in p-region





The hole well can be modified without modifying the electron barrier

Conclusion and future work

- NIPIN device
  - Symmetric and asymmetric properties
  - Number of controllable parameters
    - Width of regions
    - Doping concentration
  - Impact ionization
- Effect of using Si-Ge in i-regions

Reference

- B. Das, S. Sushama, J. Schulze and U. Ganguly, "Sub-0.2 V Impact Ionization in Si n-i-p-i-n Diode," in *IEEE Transactions on Electron Devices*
- R. Meshram *et al*., "High performance triangular barrier engineered NIPIN selector for bipolar RRAM," *2014 IEEE 6th International Memory Workshop (IMW)*